Window Function

1.Row_number():

```
SELECT emp_name,
    department,
    salary,
    ROW_NUMBER() OVER (
      PARTITION BY department
      ORDER BY salary DESC
    ) AS row_num
FROM Employee;
```

ename	department	salary	row_num
raghav	Finance	35000	1
Riya	Finance	25000	2
Rohan	Finance	22000	3
Priya	HR	18000	1
Ram	HR	15000	2
John	HR	10000	3
samu	IT	26000	1
Siya	IT	20000	2
Neha	Marketing	16000	1
Shyam	Marketing	8000	2

2.Rank():

```
SELECT ename,
    salary,
    RANK() OVER (ORDER BY salary DESC) AS rank_no
FROM employee;
```

ename	salary	rank_no
raghav	35000	1
samu	26000	2
Riya	25000	3
Rohan	22000	4
Siya	20000	5
Priya	18000	6
Neha	16000	7
Ram	15000	8
John	10000	9
Shyam	8000	10

3.DENSE_RANK():

```
SELECT
   ename,
   salary,
   DENSE_RANK() OVER (ORDER BY salary DESC) dr
FROM employee;
```

ename	salary	dr
raghav	35000	1
samu	26000	2
Riya	25000	3
Rohan	22000	4
Siya	20000	5
Priya	18000	6
Neha	16000	7
Ram	15000	8
John	10000	9
Shyam	8000	10

4.LAG():

SELECT emp_name,

salary,

LAG(salary) OVER (ORDER BY salary) AS previous_salary

FROM Employee;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ename	salary	previous_salary
Shyam	8000	NULL
John	10000	8000
Ram	15000	10000
Neha	16000	15000
Priya	18000	16000
Siya	20000	18000
Rohan	22000	20000
Riya	25000	22000
samu	26000	25000
raghav	35000	26000

5.LEAD():

SELECT emp_name,

salary,

LEAD(salary) OVER (ORDER BY salary) AS next_salary

FROM Employee;

Result Grid Filter Rows: Export: Wrap Cell Content:

ename	salary	next_salary
Shyam	8000	10000
John	10000	15000
Ram	15000	16000
Neha	16000	18000
Priya	18000	20000
Siya	20000	22000
Rohan	22000	25000
Riya	25000	26000
samu	26000	35000
raghav	35000	NULL

6.TOTAL():

```
SELECT
  ename,
  salary,
  SUM(salary) OVER (ORDER BY id) AS running_total
FROM employee;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

ename	salary	running_total
John	10000	10000
Siya	20000	30000
Ram	15000	45000
Riya	25000	70000
Shyam	8000	78000
Priya	18000	96000
Rohan	22000	118000
Neha	16000	134000
samu	26000	160000
raghav	35000	195000

7. Average():

```sql
SELECT emp_name,
    department,
    salary,
    AVG(salary) OVER (PARTITION BY department) AS avg_salary
FROM Employee;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

ename	salary	running_total
John	10000	10000
Siya	20000	30000
Ram	15000	45000
Riya	25000	70000
Shyam	8000	78000
Priya	18000	96000
Rohan	22000	118000
Neha	16000	134000
samu	26000	160000
raghav	35000	195000

8.FIRST_VALUE():

```sql
SELECT
   ename,
   salary,
   FIRST_VALUE(salary) OVER (ORDER BY salary) AS lowest_salary
FROM employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ename	salary	lowest_salary
Shyam	8000	8000
John	10000	8000
Ram	15000	8000
Neha	16000	8000
Priya	18000	8000
Siya	20000	8000
Rohan	22000	8000
Riya	25000	8000
samu	26000	8000
raghav	35000	8000

9. LAST_VALUE():

```
SELECT
  ename,
  salary,
  LAST_VALUE(salary) OVER (
    ORDER BY salary
    ROWS BETWEEN UNBOUNDED PRECEDING
    AND UNBOUNDED FOLLOWING
  ) AS highest_salary
FROM employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ename	salary	highest_salary
Shyam	8000	35000
John	10000	35000
Ram	15000	35000
Neha	16000	35000
Priya	18000	35000
Siya	20000	35000
Rohan	22000	35000
Riya	25000	35000
samu	26000	35000
raghav	35000	35000

APPLY VIEW ON DATA:

```
CREATE VIEW employee_view AS
SELECT
  id,
  ename,
  salary
FROM employee;
```

APPLY TRIGGER ON DATA;

```
DELIMITER $$

CREATE TRIGGER check_salary
BEFORE INSERT
ON employee
FOR EACH ROW
BEGIN
  IF NEW.salary < 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Salary cannot be negative';
  END IF;
END$$

DELIMITER ;
```

Handle case when in the data:

```
SELECT
  ename,
  salary,
  CASE
    WHEN salary >= 60000 THEN 'High Salary'
    WHEN salary >= 40000 THEN 'Medium Salary'
    ELSE 'Low Salary'
  END AS salary_category
FROM employee;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

ename	salary	salary_category
John	10000	Low Salary
Siya	20000	Low Salary
Ram	15000	Low Salary
Riya	25000	Medium Salary
Shyam	8000	Low Salary
Priya	18000	Low Salary
Rohan	22000	Low Salary
Neha	16000	Low Salary
samu	26000	Medium Salary
raghav	35000	High Salary